Tomato TDS Guide

What TDS is, the best levels for tomatoes, and how to fix problems

Quick answer: TDS (total dissolved solids) shows how much salt and nutrient is dissolved in water. For tomatoes, a nutrient solution is usually about **1000 to 1750 ppm**, depending on the growth stage (values on a 500 scale). Irrigation *water* itself should be much lower, ideally under about 500 ppm. TDS shows the total amount only. It does not tell you which nutrients are present.

1. What is TDS?

TDS is measured in ppm (parts per million), which is the same as mg/L. A TDS meter tells you the total amount of dissolved salts and nutrients in the water. A higher number means more dissolved solids.

Important: TDS meters use different conversion scales (usually 500 or 700). The same water can read 1000 ppm on a 500-scale meter and 1400 ppm on a 700-scale meter. All numbers in this guide use the **500 scale**. If your meter uses the 700 scale, multiply the numbers here by 1.4.

2. Best levels for tomatoes

Stage	TDS (500 scale)	TDS (700 scale)
Seedlings	500 to 750 ppm	700 to 1050 ppm
Vegetative growth	1000 to 1250 ppm	1400 to 1750 ppm
Flowering and fruiting	1250 to 1750 ppm	1750 to 2450 ppm
Source irrigation water (before adding fertilizer)	Below about 500 ppm	Below about 700 ppm

- Slightly higher TDS (about 1750 to 2500 ppm on the 500 scale) can make fruit sweeter and tastier, but it lowers fruit size and total yield.
- Tomatoes are moderately sensitive to salt. Yield falls as the salt level in the soil or solution rises.
- Hot weather increases water use, so salts build up faster. Use a lower TDS in hot weather and a higher TDS in cool weather.

3. Signs of wrong TDS

Level	Signs	Common causes
Too high	Dark green, stiff and small leaves; burnt leaf edges; wilting even when the soil is wet; blossom end rot; small fruit; slow growth	Too much fertilizer; salty water; not enough watering to wash out salts; hot, dry weather
Too low	Pale or yellow leaves, thin stems, slow growth, few flowers	Too little fertilizer; heavy rain or over-watering that washes nutrients out; very diluted solution

Yellow leaves: both too low TDS (hungry plant) and too high TDS (salt stress, damaged roots) can cause yellow leaves. Measure before you add more fertilizer.

4. How to measure

- Calibrate your meter often with a standard calibration solution, and follow the meter instructions.
- Measure at about 25 °C. Use a meter with temperature compensation, because warm water gives a higher reading.
- For hydroponics, measure the tank solution. For soil or substrate, also measure the **runoff** (the water that drains out). Runoff much higher than the water you applied means salts are building up.
- Measure at the same time every day so you can compare readings.

5. How to fix it

Problem	What to do
TDS too high	Flush the soil or substrate with clean, low-TDS water (leach about 10 to 20% extra water each time you irrigate). Dilute the nutrient solution. Reduce the fertilizer amount. Improve drainage. Use mulch to reduce evaporation.
TDS too low	Increase the fertilizer in small steps (about 100 to 150 ppm at a time) and measure again. Check that nutrients are balanced, not only the total.
Salty irrigation water	Test the water source. Mix with better water or rainwater, use drip irrigation with leaching, and choose salt-tolerant varieties or rootstocks.

6. Simple checklist

- Test your source water first. Subtract its TDS when you plan the fertilizer.
- Set a target TDS for the growth stage and measure at least once a day in hydroponics.
- Check runoff in soil or substrate every week.
- Change the TDS slowly, never by a big jump.
- If the plants still look wrong, do a lab test for the individual nutrients, sodium and chloride.

Note: The numbers are typical ranges. The best TDS depends on the variety, climate, growing system and fertilizer. Follow your fertilizer label and ask a local agronomist for advice.